Jaewoo Kim

Ph.D. Candidate, Department of Aerospace Engineering

Korea Advanced Institute of Science and Technology (KAIST)

291 Daehak-ro, Yuseong, Daejeon 34141, Republic of Korea

Phone: +82-42-350-3770 | Mail: jw.kim@kaist.ac.kr | Linkedin | Website

RESEARCH INTERESTS

Space Systems Design & Operations

- Lifecycle-based space systems design considering stakeholder interactions and operational constraints
- Planning and management of space operations for efficient, resilient, and sustainable space utilization

Engineering Decision-Making Problems

- Formulation of complex engineering problems by identifying key system drivers
- Quantitative decision-making frameworks for interpretable and actionable engineering insights

EDUCATION

Korea Advanced Institute of Science & Technology (KAIST) — Daejeon, Korea

Ph.D. in Aerospace Engineering — *Feb. 2027 (Expected)*

- Thesis (in progress): Maintenance Strategies for Satellite Mega-Constellations (Advisor: Prof. Jamyung Ahn)

M.S. in Aerospace Engineering — *Feb. 2024*

- Thesis: Optimal Satellite System Architecting Considering On-Orbit Refueling (Advisor: Prof. Jamyung Ahn)

Seoul National University (SNU) — Seoul, Korea

B.S. in Mechanical and Aerospace Engineering — *Feb. 2022*

- Thesis: Celestial Navigation Using Stars and Planets on Lunar Exploration Orbit (Advisor: Prof. Changdon Kee)

PUBLICATIONS

Journal Articles (Corresponding Author Underlined)

[J1] **Kim, J.**, Sung, T., Hwang, W., and Ahn, J., "On-Orbit Servicing-Integrated Maintenance Strategy for Satellite Constellation," accepted by *AIAA Journal of Spacecraft and Rockets*.
doi:10.48550/arXiv.2512.18985

[J2] **Kim, J.** and Ahn, J., "System Architecting for GEO Communication Satellite Considering On-Orbit Refueling," *AIAA Journal of Spacecraft and Rockets*, Vol. 63, No. 1, 2026, pp. 176–191.
doi:10.2514/1.A36403

[J3] **Kim, J.**, Choi, E., Ahn, J., Suh, E. S., Kim, J.-H., and Lim, D. G., "Mathematical Programming-Based Design Structure Matrix Clustering for Optimal Modular Architecture Design," *ASME Journal of Mechanical Design*, Vol. 147, No. 10, 2025, pp. 101401.
doi:10.1115/1.4068100

[J4] **Kim, J.**, Ahn, J., and Sung, T., "Replenishment Strategy for Satellite Constellation with Dual Supply Modes," *AIAA Journal of Spacecraft and Rockets*, Vol. 62, No. 5, 2025, pp. 1576–1583.
doi:10.2514/1.A36281

[J5] Ko, J., **Kim, J.**, Choi, J., and Ahn, J., "Simultaneous Optimization of Launch Vehicle Stage and Trajectory Considering Flight-Requirement Constraints," *KSAS International Journal of Aeronautical and Space Sciences*, Vol. 25, 2024,

pp. 1563–1573.
doi:10.1007/s42405-024-00737-1

Papers Written in Korean

[K1] **Kim J.**, Lee, J., Choi, E. J., Choi, J., Yu, J., Jo, J., Kam, H., and Ahn, J., "Development of 3D Cell Model for Collision Risk Assessment of Space Assets," *KSSS Journal of Space Technology and Applications*, Vol. 25, No. 2, 2025, pp. 114–124.
doi:10.52912/jsta.2025.5.2.114

Preprint & Work in Progress

[P1] Choi, E., **Kim, J.**, Kim, J.-H., Suh, E. S., and Ahn, J., "Optimal Design Structure Matrix Clustering with Constraint Selections for Conflict Resolution," under review at *ASME Journal of Mechanical Design*.

[P2] Park, B., Howell, K. C., **Kim, J.**, and Ahn, J., "Families of Two-Impulse Optimal Rendezvous Transfers Between Elliptic Orbits," under revision review at *AAS Journal of the Astronautical Sciences*.
doi:10.48550/arXiv.2603.11538

[P3] **Kim, J.**, Song, M., Lee, J., Yu, J., Choi, E. J., Kam, H., Jo, J., and Ahn, J., "Korean Space Collision Risk Analysis Software Based on 3D-Cell Model," under review at *KSAS International Journal of Aeronautical and Space Sciences*.

[P4] **Kim, J.**, Sung, T., Hwang, W., and Ahn, J., "Joint Replenishment Strategy for Multiple Satellite Constellations with Shared Launch Opportunities," in preparation.
doi:10.48550/arXiv.2503.23666

[P5] **Kim, J.**, Choi, E., Suh, E. S., Kim, J.-H., and Ahn, J., "Managing Module Boundaries Across the OEM-Supplier Interface: A Bi-Objective DSM Clustering Approach," in preparation.

[P6] Lee, D., Ahn, J., and **Kim, J.**, "Multi-Domain Matrix Framework for Human Resource Decision Support," in preparation.

[P7] **Kim, J.**, Lee, S., Choi, E., Kim, J.-H., and Ahn, J., "Constrained Modularity Maximization in Complex Networks via Graph Partitining and Column Generation," in preparation.

[P8] **Kim, J.**, Ahn, J., and Choi, J., "Reinforcement Learning-Based Architecture Design Optimization of Large Space Systems with On-Orbit Servicing," in preparation.

[P9] **Kim, J.**, Song, M., Yu, J., Choi, E. J., Kam, H., Jo, J., and Ahn, J., "Cooperative Ground and Space Sensors Tasking for Emergency Collision Events," in preparation.

Proceedings of Refereed Conferences

[C1] Lee, S., **Kim, J.**, and Ahn, J., "A Multi-Trip Time-Dependent Traveling Salesman Problem for Space Logistics," in *2026 INFORMS Annual Meeting*, San Francisco, California, US, Nov. 1–4, 2026. (to appear)

[C2] **Kim, J.**, Song, M., Lee, J., Yu, J., Kam, H., Choi, E.-J., Choi, J., Jo, J. H., and Ahn, J., "Inclination-Aware 2D Source-Sink Modeling for Long-Term Projection of the LEO Collision Environment," in *27th Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference*, Maui, Hawaii, US, Sep. 15–18, 2026. (to appear)

[C3] Song, M., **Kim, J.**, and Ahn, J., "Decision Support for Space Debris Remediation Enterprises via Two-Stage System Optimization under Various Future Scenarios," in *27th Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference*, Maui, Hawaii, US, Sep. 15–18, 2026. (to appear)

[C4] **Kim, J.**, Choi, J., and Ahn, J., "Flexibility-Aware Assessment and Optimization of System Architectural Decision for Large Space Systems Considering On-Orbit Servicing," in *Asian Congress of Structural and Multidisciplinary Optimization 2026*, Busan, Korea, May 17–21, 2026.

[C5] **Kim, J.**, Sung, T., Hwang, W., and Ahn, J., "Integration of On-Orbit Servicing into Constellation Maintenance with Spare Inventory Management," in *2026 AIAA SciTech Forum*, Orlando, Florida, US, Jan. 12–16, 2026.
doi:10.2514/6.2026-2631

[C6] Song, M., **Kim, J.**, Nam, Y., and Ahn, J., "Low Earth Orbit Space Debris Remediation Program for Maximizing Stakeholders' Utility," in *2025 Asia-Pacific International Symposium on Aerospace Technology (APISAT)*, Seoul, Korea, Oct. 27–29, 2025.

[C7] **Kim, J.**, Sung, T., and Ahn, J., "Joint Replenishment Strategy for Multiple Satellite Constellations," in *2025 AIAA SciTech Forum*, Orlando, Florida, US, Jan. 6–10, 2025.
doi:10.2514/6.2025-1757

[C8] **Kim, J.**, Lee, J., Kim, H., Choi, E. J., Choi, J., Yu, J., Jo, J., and Ahn, J., "Development of Korea Orbital Debris Evolutionary and Engineering Model," in *75th International Astronautical Congress*, Milan, Italy, Oct. 14–18, 2024.
doi:10.52202/078360-0161

Proceedings Written in Korean

[C9] **Kim, J.**, Ahn, J., Park, B., and Howell, K. C., "A Family-Based Framework for Two-Impulse Optimal Rendezvous Transfers," in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Busan, Korea, Jun. 24–26, 2026.

[C10] **Kim, J.**, Song, M., Yu, J., Kam, H., Choi, E. J., Choi, J., and Jo, J. H., and Ahn, J., "3D Cell Model-Based Collision Risk Analysis for Historical and Prospective LEO Scenarios," in *Korean Academy of Space Security (KASS) Spring Conference*, Yeosu, Korea, May 14–15, 2026.

[C11] **Kim, J.**, Song, M., Choi, E. J., Choi, J., Yu, J., Jo, J., Kam, H., and Ahn, J., "Updates on Space Collision Risk Analysis Framework," in *Korean Space Science Society (KSSS) Spring Conference*, Yeosu, Korea, Apr. 22–24, 2026.

[C12] **Kim, J.**, Yi, G., and Ahn, J., "Implications of In-situ Resource Utilization from Space Logistics Perspective," in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Mar. 31–Apr. 3, 2026.

[C13] **Kim, J.**, Yi, G., and Ahn, J., "Research and Standardization of Space Logistics for Long-term Complex Space Mission," in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Mar. 31–Apr. 3, 2026.

[C14] **Kim, J.**, Sung, T., Song, M., and Ahn, J., "Integrated Inventory Model for Constellation Maintenance via Spare Satellites and On-Orbit Servicing," in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Goseong, Korea, Nov. 11–14, 2025.

[C15] **Kim, J.**, Song, M., and Ahn, J., "Optimal Design Lifetime and Propellant Loading of GEO Communication Satellites Considering On-Orbit Refueling," in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Goseong, Korea, Nov. 11–14, 2025.

[C16] **Kim, J.**, Choi, E. J., Choi, J., Yu, J., Jo, J., Kam, H., and Ahn, J., "Survey of Risk Indices for Space Objects and Implications," in *Korean Space Science Society (KSSS) Fall Conference*, Jeju, Korea, Oct. 29–Nov. 1, 2025.

[C17] Lee, J., Lee, J., Jang, E., Jeon, H., **Kim, J.**, Lee, C., Scheeres, D. J., and Han, J., "Asteroid Accessibility Searching Algorithm," in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Yeosu, Korea, Jun. 25–27, 2025.

[C18] **Kim, J.**, Song, M., and Ahn, J., "Survey of Research on Economic and Performance Analysis of On-Orbit Servicing and Implications," in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Yeosu, Korea, Jun. 25–27, 2025.

[C19] **Kim, J.**, Ahn, J., Choi, E. J., Choi, J., Yu, J., and Jo, J., "Development of Integrated Space Collision Risk Response System," in *Korean Academy of Space Security (KASS) Spring Conference*, Seoul, Korea, May 9, 2025.

[C20] **Kim, J.**, Choi, E. J., Choi, J., Yu, J., Jo, J., and Ahn, J., "Progress of Space Debris Environment and Risk Analysis Framework Development," in *Korean Space Science Society (KSSS) Spring Conference*, Seoul, Korea, Apr. 22–24, 2025.

[C21] **Kim, J.**, Sung, T., and Ahn, J., "Application of Space Logistics: Maintenance Strategy of Satellite Mega-Constellation," in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Apr. 2–4, 2025.

[C22] **Kim, J.**, Lee, J., Choi, E. J., Jin, C., Yu, J., Jo, J., and Ahn, J., "Development of Korean 3D Cell Model for Space Debris Environment Analysis," in *Korean Space Science Society (KSSS) Fall Conference*, Sacheon, Korea, Oct. 28–30, 2024.

[C23] **Kim, J.** and Ahn, J., "An Integrated Inventory Management Model for Maintenance of Multiple Satellite Constellations," in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Changwon, Korea, Jun. 26–28, 2024.

[C24] **Kim, J.**, Ko, J., Choi, J., Ahn, J., Yoon, N., and Kim, H., "Conceptual Design of Launch Vehicle Considering Axial Acceleration Constraints," in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Changwon, Korea, Jun. 26–28, 2024.

[C25] Ko, J., **Kim, J.**, Choi, J., Ahn, J., Yoon, N., and Kim, H., "Development of Conceptual Design Software for Space Launch Vehicle," in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Apr. 3–5, 2024.

[C26] **Kim, J.**, and Ahn, J., "Multiobjective Design Optimization of Commercial Satellite Considering On-Orbit Refueling Policy," in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Apr. 19–21, 2023.

[C27] **Kim, J.**, Lee. D. U., and Ahn, J., "Research on the Overseas On-Orbit Servicing Trends and Implications," in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Jeju, Korea, Nov. 16–18, 2022.

RESEARCH EXPERIENCE

Graduate Research Assistant | *Strategic Aerospace Initiative, KAIST* *Mar. 2022 – Present*

- **In-Space Servicing and Manufacturing Research Center (ISMRC)**
  Duration: 2025–2035; Sponsor: National Research Foundation of Korea
  - Developed a maintenance framework for satellite mega-constellations integrating on-orbit servicing [C5]
  - Surveyed standardization of space logistics framework [C12, C13]
- **Space Logistics Modeling and Demand Fulfillment Strategy Evaluation Framework**
  Duration: 2025–2028; Sponsor: National Research Foundation of Korea
  - Developed frameworks for evaluating the value of on-orbit servicing and potential architectural changes for GEO communication satellite systems [J2, C4, C15, C18]
  - Developed maintenance strategies for satellite mega-constellations [J1, P4, C5, C7, C14, C21]
  - Supported conceptual design for a space debris remediation program [C3, C6]
  - Surveyed in-situ resource utilization from logistics perspective [C12]
- **Multi-Criteria Decision Making for Optimal Modular System Architecture Selection**
  Duration: 2024–2026; Sponsor: Hyundai Motor Company
  - Surveyed multi-criteria decision-making methods and their applications in system architecting
  - Developed a mathematical programming approach for design structure matrix-based modular architecting with make-buy decision [P5]
  - Supported the development of a mathematical programming approach for modular architecting with conflict resolution among design constraints [P1]
  - Developed a mathematical programming-based framework for constrained modularity maximization [P7]
- **Development of Risk Analysis Framework for Korea Space Situational Awareness**
  Duration: 2024–2027; Sponsor: Korea Astronomy and Space Science Institute

- Surveyed models for the space object environment and collision-risk analysis
- Developed a Korean orbital debris engineering model [P3, K1, C8, C10, C11, C16, C19, C20, C22]
- Developed a ground-space cooperative sensor tasking framework for operational planning [P9]

- **A Study on the Principle of Modular Architecture Engineering to Improve Level of Completion for Vehicle Architecture**
  Duration: 2022–2023; Sponsor: Hyundai Motor Company
  - Developed a mathematical programming approach for design structure matrix-based modular architecting [J3]
  - Conducted case studies on automobile subsystems and identified improved design solutions
- **Research on ADR/OOS Applications for National Security Space Assets**
  Duration: 2022; Sponsor: Korean Society for Aeronautical and Space Sciences
  - Reviewed on-orbit servicing technologies and related mission concepts [C27]
  - Developed concepts of operations for ADR/OOS missions
- **Development of Launch Vehicle Mission & Conceptual Design Software**
  Duration: 2022–2023; Sponsor: Hanwha Aerospace
  - Developed propulsion and staging modules for multidisciplinary design optimization
  - Contributed to the development of an all-at-once design optimization framework considering flight requirements [J5, C24, C25]

Undergraduate Researcher | *GNSS Laboratory, SNU* — *Mar. 2021 – Aug. 2021*

- **Deep Space Navigation with Optical Sensor Data**
  - Reviewed non-inertial deep-space navigation algorithms
  - Evaluated the selected algorithm through numerical simulations

TEACHING EXPERIENCE

Teaching Assistant | *KAIST* — *Sep. 2023 – Present*

- AE210 Aerospace Thermodynamics – Spring 2024
- AE280 Software Application in Aerospace Engineering – Spring 2026, Spring 2025
- AE401 Aerospace System Design II – Fall 2023
- AE450 Flight Dynamics and Control – Fall 2024
- AE551 Introduction to Optimal Control – Fall 2025

Part-Time Lecturer | *Humaiin co.* — *Aug. 2022 – Present*

- Delivered lectures and developed instructional materials covering foundational data science concepts, programming tools, and LLM-based workflow automation.
- Institutions: Busan City Government, Korea Education & Research Information Service (KERIS), Statistics Korea (KOSTAT), Ewha Womans University, Sookmyung Women's University, Seoul AI Foundation, Hanwha Solutions, Lotte Chemical

PROFESSIONAL SERVICES

- Reviewer for *INCOSE Systems Engineering*; *AIAA SciTech Forum*, *AIAA AVIATION Forum*.

AWARD & HONORS

2025 Korean Space Science Society Spring Conference | *Best Oral Presentation Award* — *Apr. 2025*

- Presentation Title: Progress of space debris environment and risk analysis framework development

Optimization Grand Challenge 2024 | *Top 10% of Participants* *Oct. 2024*

- Competition to develop an optimization algorithm for pickup-and-delivery problems, sponsored by LG CNS
- Team Name: J2Opt

Hanhwa-KAIST Space Hub Space Grand Challenge | *Bronze* *Nov. 2023*

- Competition for innovative space systems, technologies, and mission concepts, sponsored by Hanhwa Aerospace
- Team Name: LETA (Lunar Exploration Trajectory Analytics)
- Topic: Lunar exploration trajectory design with low-thrust propulsion and multiple gravity assist
- Prize: 3,000,000 KRW (2,050 USD)

Extracurricular Experience

Military Service | *Defense Security Command (DSC)* *Apr. 2018 – Nov. 2019*

- Supported educational programs for military officers in DSC, and served as a squad leader
- Received commendation from DSC School Head (Brigadier General)

Interviewer | *Humans of SNU* *Jul. 2017 – Dec. 2017*

- Interviewed diverse members of the SNU community and discovered insightful and interesting stories

Volunteer | *People to People International* *Mar. 2016 – Feb. 2018*

- Supported underprivileged members in Seoul City
- Student chapter union president (2017); SNU chapter president (2016)

Other Skills

Programming

- Python, MATLAB, Julia, C, C++, Java for quantitative analysis, including optimization, simulation, and statistical reasoning

Language

- Korean (native), English